AI Engineer

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Evaluation, monitoring, and oversight of deployed AI systems

Staff-level AI and full-stack engineer with 10 years building and evaluating AI systems in production across language, voice, vision, and robotics. Most recently tech lead for LLM voice agents handling ~50k calls a day, where I built the evaluation and production-monitoring system. Moving into AI safety research, focused on scalable oversight and AI control.

Research & AI Safety

- **Publication:** *Mean Shift Rejection: Training Deep Neural Networks Without Minibatch Statistics or Normalization* (arXiv, 2019) — second author. Studied an alternative to batch normalization in ResNets; faster convergence with no loss in accuracy. Code on GitHub.
- **Independent control experiments (2026):**
  - *Agent swarm contagion:* measure whether one rule-breaking agent spreads eval-gaming behavior through a swarm of honest agents on a shared board.
- **AI safety research group (Oct 2023 – Apr 2024):** weekly group with alignment researchers focused on LLM consistency and control; read and gave feedback on papers on output monitoring and truthfulness.
- **BlueDot Impact AI safety course (Mar 2025):** intensive course of daily sessions.

Experience

Staff AI Engineer, Tech Lead *Hiya* — 05/2025 - 05/2026

LLM Voice Agents **For businesses · ~50k production calls/day**

- Tech lead building a platform for LLM voice agents making and answering calls for businesses; owned technical direction end to end across infrastructure, backend, and frontend, met with customers, and shaped product vision
- Built the evaluation system from scratch: hundreds of simulated call scenarios and assertions per agent, graded by LLM judges and run daily across every agent to catch regressions
- Extended the same judges to production as a post-call monitor, grading live calls after they ended and surfacing failures for human review, providing oversight at a scale no human team could cover
- Connected judges to agent-harness signals to catch faulty tool calls, such as failing to transfer a caller to a human when required
- Built a failure taxonomy using axial coding and a human review UI to label and triage failure types

Cofounder, CTO *Of Counsel AI* — 07/2023 - 09/2024

Legal AI **Copilot**

- Built an LLM evaluation framework with custom datasets, enabling data-driven model selection and a 25% quality improvement
- Architected a production RAG system combining graph databases and LLMs for legal drafting and analysis
- Built an agent system with distributed workflow execution for multi-step document processing, increasing batch speed 80%
- Built a copilot integrating chat, document editing, and UI actions, on a real-time collaborative editor (CRDTs)
- Selected for Madrona Venture Labs' Launchable startup competition

Founder *Weaver.fm* — 09/2022 - 04/2023

Social Audio App

- Built a custom ASR pipeline combining Whisper with forced alignment, reducing transcription errors 40%
- Used LLM-generated prompts to facilitate conversations, tripling average conversation length

SDE, Robotics Simulation *Amazon Scout* — 07/2020 - 08/2022

Autonomous Delivery **Sidewalk delivery robots**

- Built a multimodal autoencoder and clustering system to identify and classify incidents, discovering new robot failure modes
- Led development of a GPU-accelerated cloud simulation platform running 5,000+ photorealistic, physics-based tests daily against new robot software versions

- Built a pipeline generating 1M+ synthetic sensor training samples monthly for navigation and perception models
- Raised GPU utilization 25% and cut job queue times 50% through multi-region autoscaling
- Architected software distribution with rollback for 1,000+ physical robot deployments weekly

Machine Learning Engineer *AI Foundation* 01/2019 - 02/2020

Generative AI **Avatars**

- Developed a novel alternative to batch normalization that converged faster without losing accuracy (see Publication)
- Led training and deployment of ML models across the engineering organization, including a voice-cloning pipeline
- Ran ablation studies of embedding models and built a semantic similarity service, improving response accuracy 30%

SDE, Machine Learning Services *Amazon* 08/2016 - 01/2019

Alexa Shopping

- Founding engineer on the Alexa Shopping ML team; built language model infrastructure serving 40M+ monthly users
- Designed key components of data annotation and human-in-the-loop model evaluation tools
- Built distributed inference services with 99.99% uptime and 50 ms average latency across 10M+ daily requests
- Cut model deployment time from 2 weeks to 2 days with an automated pipeline

Education

- **B.S. Computer Science, Vanderbilt University** (2016) — minor in Engineering Management; coursework in AI, Machine Learning, Neuroscience
- **Machine Learning University at Amazon** — 12 six-week courses including Probabilistic Graphical Models, Deep Learning NLP, CNNs, RNNs, Transformers
- **UC Berkeley, Data Science Masters Program** (unfinished) — coursework in Statistics and Research Design

Skills

- **ML & LLMs:** PyTorch, LLM-as-judge evaluation, agents and tool use, RAG, ASR and voice, embedding models, ablation studies
- **Languages:** Python, TypeScript, C++, CUDA, Java, Scala
- **Infrastructure:** AWS, Google Cloud, Docker, Postgres, Temporal, Neo4j, Kafka, ROS, Vulkan

Other

- **Independent consulting (2022 – 2025):** built AI prototypes for clients, including product listings generated from voice descriptions and an internal Slack Q&A agent
- **AECJ climate nonprofit (2020 – 2026):** board of directors, VP; led tech and recruiting